returns the index of the closest point, rather than the point itself. It's not hard to find a workaround if this is not the case: we can keep a hash table associating centroids to their indices, and add an extra step to retrieve the index of the centroid returned by the `KdTree`.

Listing 12.5 classifyPoints using k-d trees

```
function classifyPoints(points, centroids)
                  #1
  clusters ← []
                            #2
  kdTree ← new
KdTree(centroids)                                    #3
  for i in {0, .., |points|-1}
do                                                #4
    clusters[i] ←
kdTree.nearestNeighborIndex(point)                   #5
  return clusters
                                                          #6
```

#1 Method `classifyPoints` takes the list of points in the dataset and the current list of centroids. It returns the list of the centroids (technically, their indices) associated with each point in the dataset.
#2 Initialize the list of cluster assignments to an empty list.
#3 Create a KdTree instance, initializing it with the list of centroids.
#4 Cycle through the list of points in the dataset (their indices).
#5 Get the index of the nearest centroid for this point by querying the KdTree.
#6 Return the list of clusters.